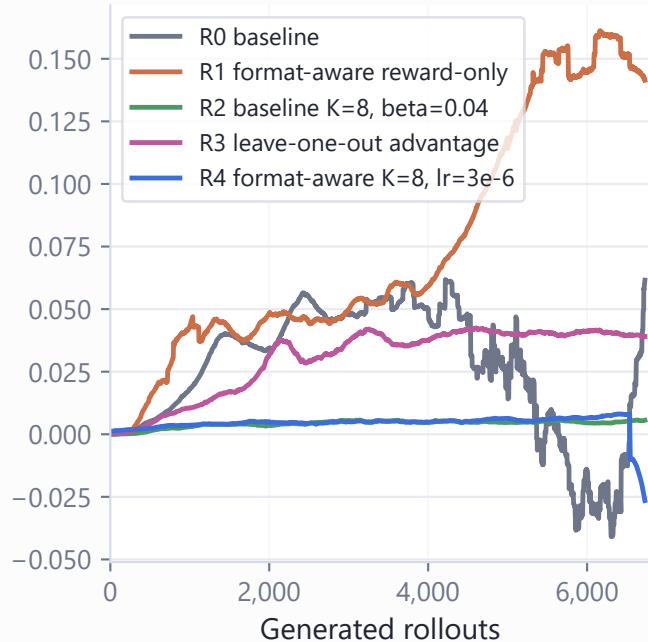


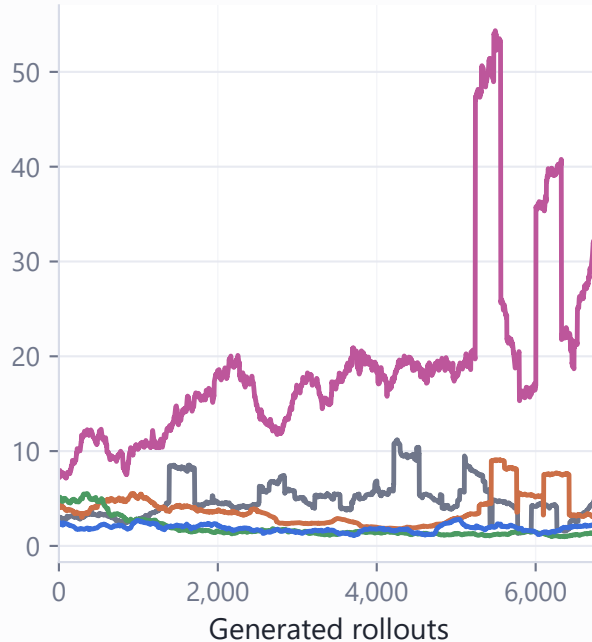
Optimization traces

Smoothed training loss, gradient norm, and actor perplexity; rollout axis is aligned across K=2 and K=8 runs.

Loss



Gradient norm



Actor perplexity

